Network Monitoring and Threat Detection Lab

Objective

The purpose of this lab was to use Wireshark to capture and analyze network traffic. The goal was to identify normal traffic and detect suspicious or attack-like behavior such as ping sweeps, port scans, and high volume ICMP traffic.

Traffic Capture Setup

Wireshark was used to capture live network traffic on the active network interface. Normal traffic was first generated using basic commands to establish a baseline for comparison.

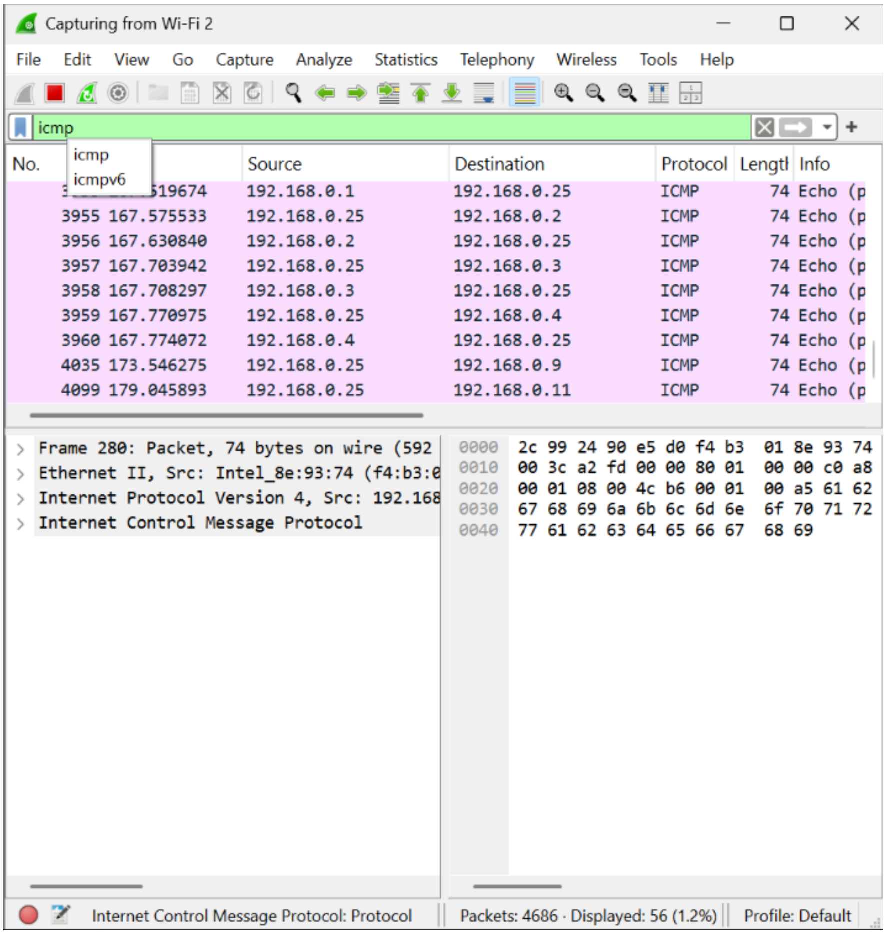
ICMP Ping Sweep

A ping sweep was performed by sending ICMP echo requests to multiple IP addresses on the local network. This simulates network scanning behavior.

In Wireshark, this was identified by:

- A single source IP sending requests
- Multiple destination IP addresses

- ICMP protocol traffic



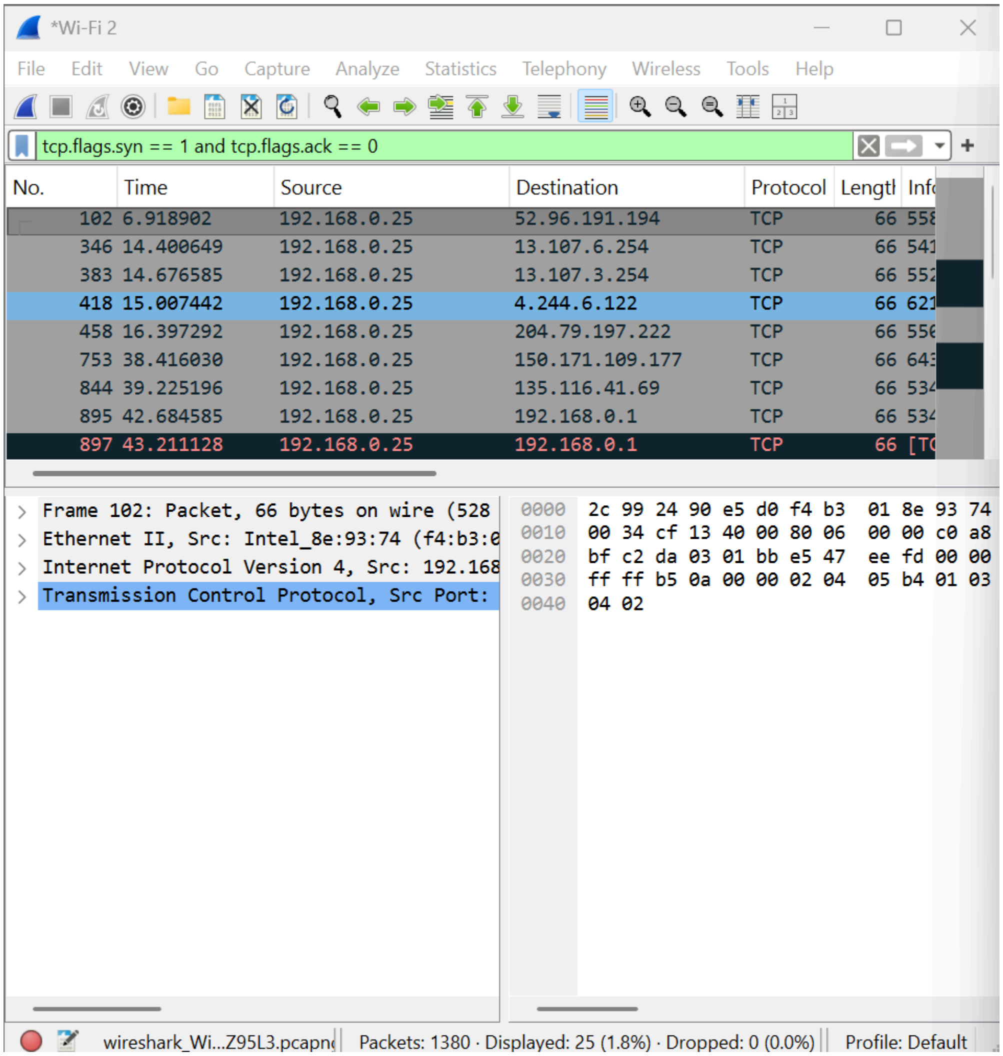
TCP SYN Scan

A port scan was simulated using PowerShell to send TCP SYN packets to multiple ports on a target device.

In Wireshark, this was identified by:

- Multiple SYN packets
- Same destination IP

- Different destination ports

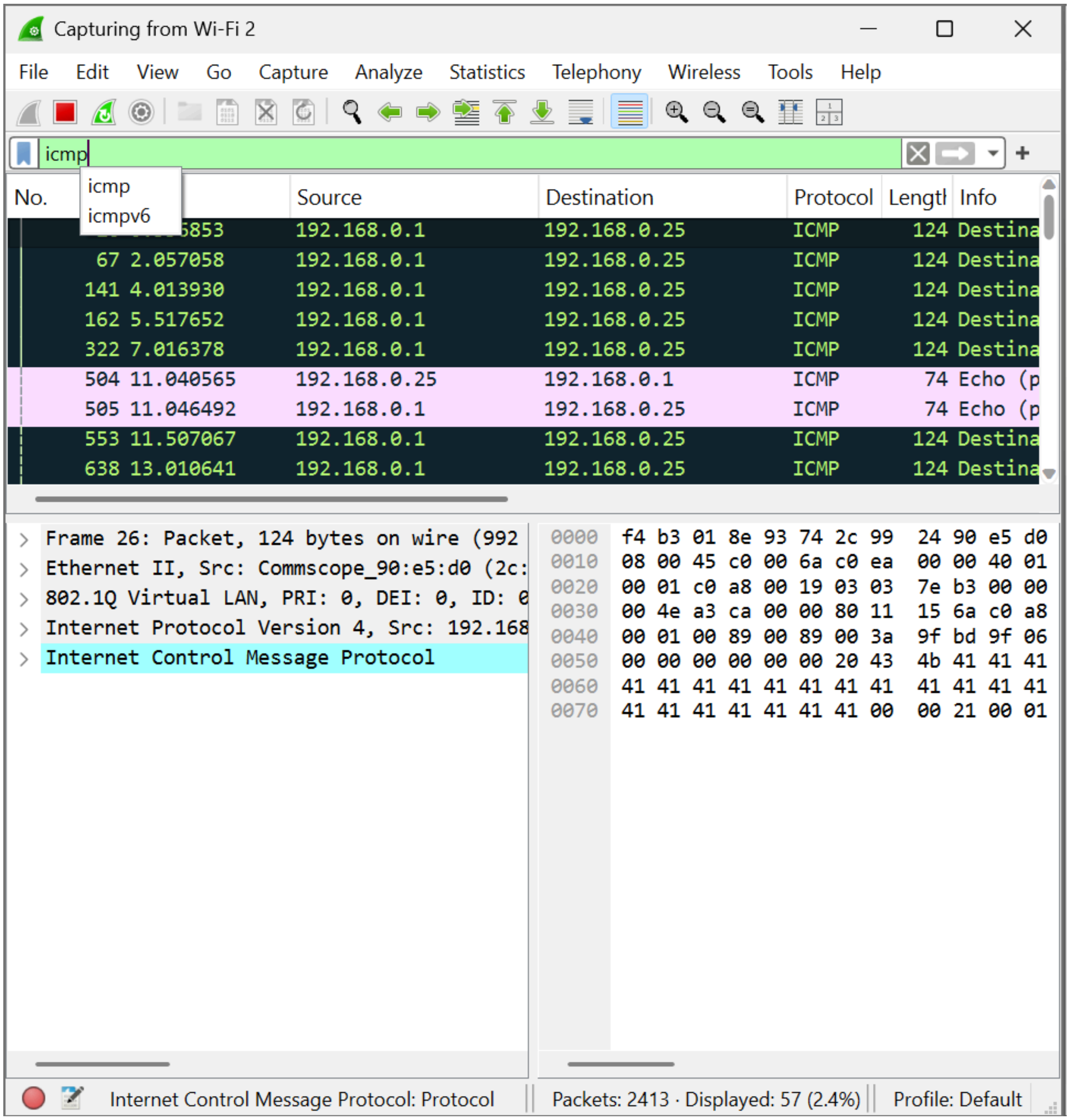


ICMP Flood (DoS-like Activity)

A high-rate ICMP flood was generated using large packet sizes. This simulates denial-of-service behavior.

In Wireshark, this was identified by:

- A large number of ICMP packets
- Repeated communication between the same devices
- Larger packet sizes than normal



Conclusion

In this lab, Wireshark was successfully used to capture and analyze both normal and suspicious network traffic. Different types of attack-like behaviors were identified, including ping sweeps, port scans, and ICMP floods. This demonstrates how network monitoring tools can be used to detect potential security threats in real time.